Pehchaan for Paytm merchants

Never lose a *regular* again.

Customer intelligence for every kirana — a churn model on the UPI payments Paytm merchants already receive, an agent that wins regulars back, and a Soundbox that speaks Kannada, Hindi and English.

Paytm Build for India AI Hackathon · Bengaluru · 31 Oct 2026
Track 1 — Merchant Growth AI + an autonomous Track 3 agent

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pehchaan-sigma.vercel.app

THE PROBLEM

Quick commerce is taking the kirana's best customers

51%

of shoppers say they rely less on their kirana than a year ago.

Grant Thornton Bharat survey, via NEXT IAS (Jul 2026)

They leave silently

Nobody cancels a kirana. They order from an app one week and simply don't come back.

400 customers, zero visibility

A shopkeeper can't notice that a regular who came every 3 days hasn't come in 3 weeks.

THE INSIGHT

Every UPI payment already carries a *name.*

Paytm sees the payer name and UPI ID on every payment across **~1.5 crore subscription merchants** — and today uses it only to announce “payment received”.

Subscription merchants at end of FY26 — Inc42

THE SOLUTION

Pehchaan: from a payment to a relationship

01

Recognise

Each Soundbox payment builds a customer profile automatically — how often, how much, which household.

02

Predict

A churn model (MBG/NBD) scores every customer's chance of still being a regular, every night.

03

Act

An agent picks who to win back, writes in Kannada, sends via the Paytm app, and confirms the return at the counter.

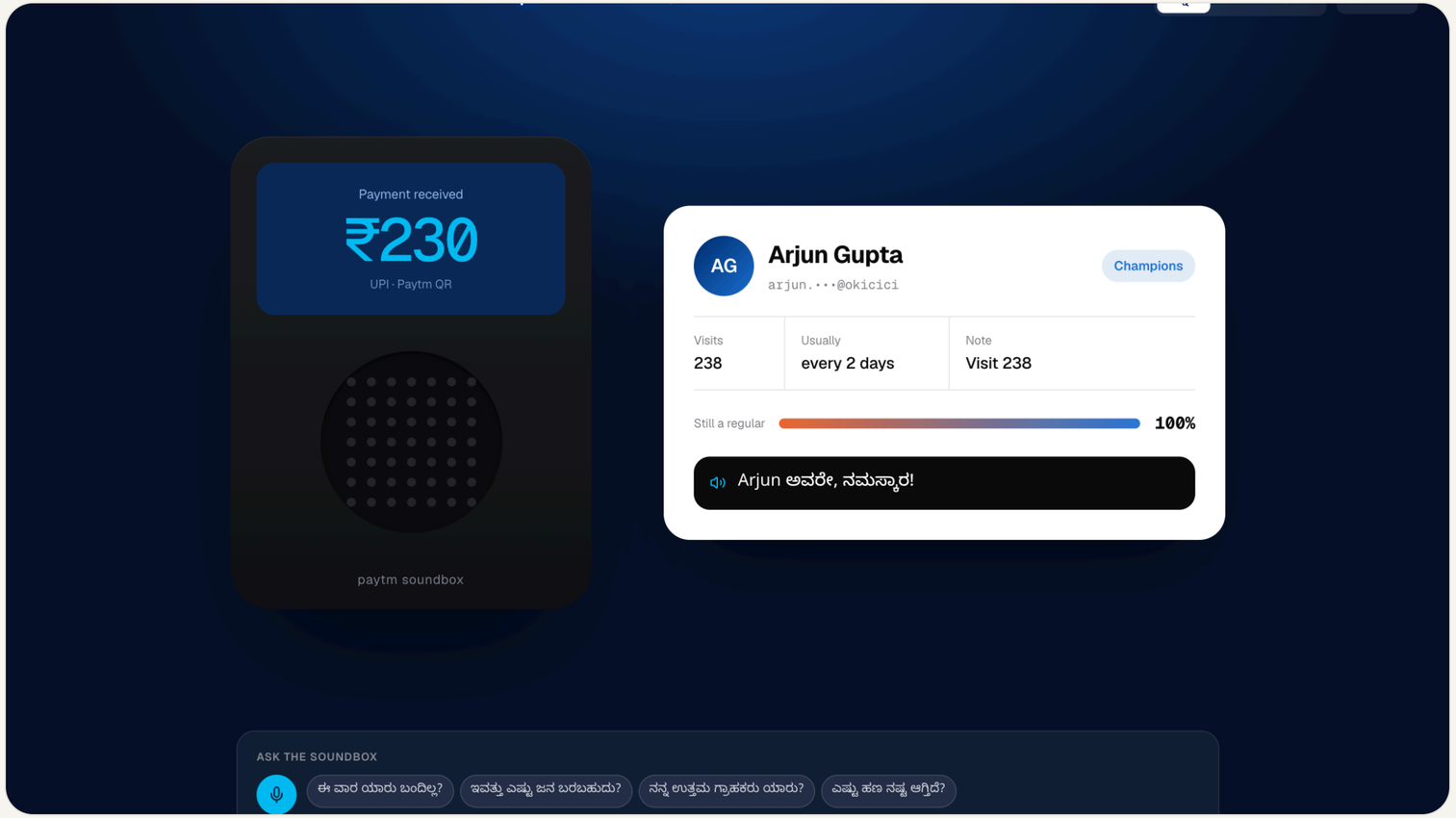
04

See the city

Across shops, Paytm spots regulars leaving the same week — and locates the new competitor.

PRODUCT · THE SOUNDBOX

One payment. One person.



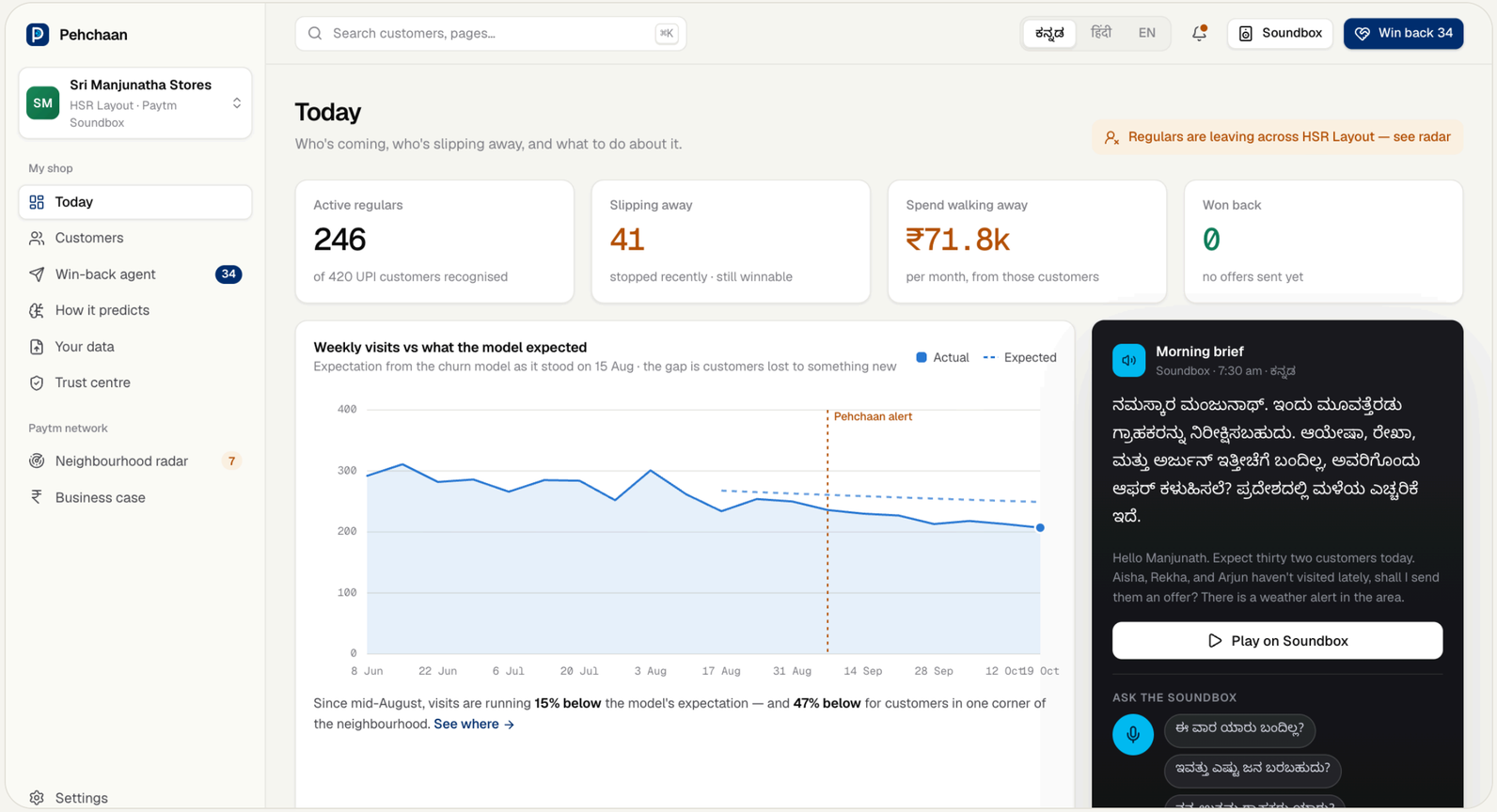
When ₹230 lands, the Soundbox knows it's **Arjun** — visit 238, every 2 days — and greets him by name.

A lapsed regular returns: “ಆಯಿಷಾ ಅವರು 48 ದಿನಗಳ ನಂತರ ಮತ್ತೆ ಬಂದಿದ್ದಾರೆ!”

The owner can ask out loud — “ಈ ವಾರ ಯಾರು ಬಂದಿಲ್ಲ?” — and hear the answer. Kannada · हिंदी · English.

PRODUCT · MERCHANT DASHBOARD

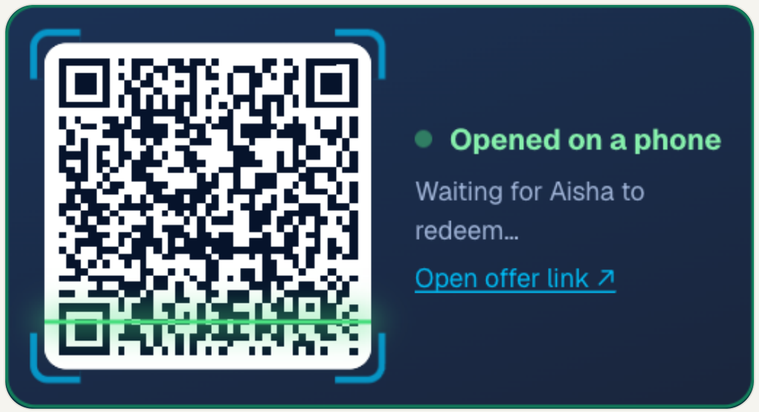
Today: who's slipping away, and what to do



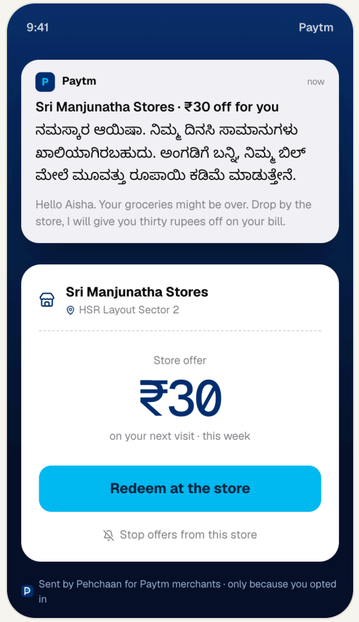
- 41 regulars slipping away — ₹72k a month in spend
- Visits vs the model's expectation; the gap is customers lost
- Morning brief and Q&A on the Soundbox

TRACK 3 · THE LOOP, END TO END

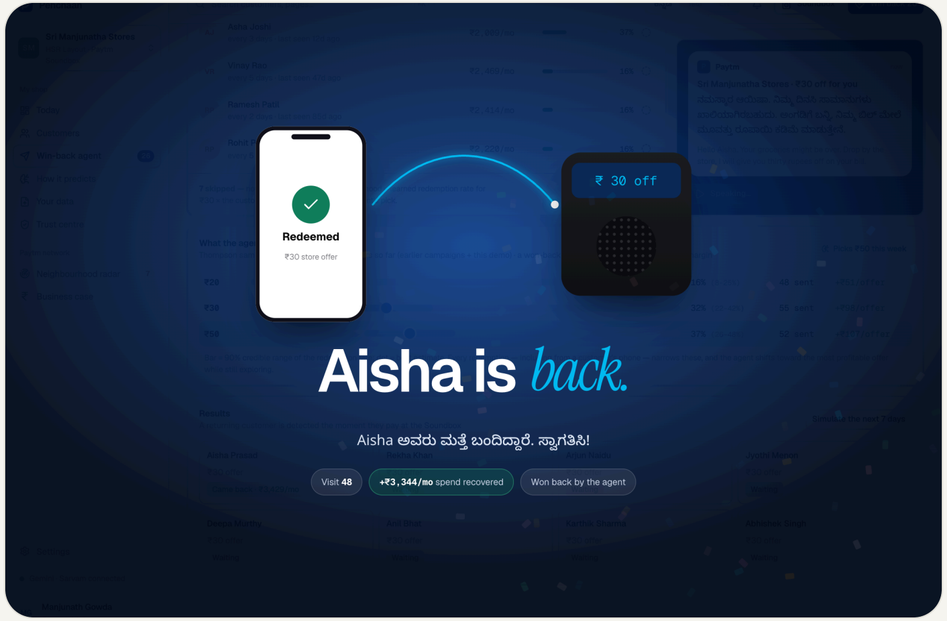
Live: scan, redeem, recognised



1 · Agent sends a QR offer



2 · Kannada offer on the phone



3 · Redeemed → “Aisha is back.”

Opened on the phone shows live on the dashboard · Redeem → the Soundbox greets her by name · production round-trip: 5 ms

HOW IT PREDICTS

Real machine learning — not an LLM wrapper

Payments

Per UPI ID: frequency, recency, customer age, spend

→

MBG/NBD

P(still a regular) and expected visits; lets one-time buyers churn

→

Gamma-Gamma

Spend per visit → 90-day customer value

→

Thompson sampling

Learns which offer (₹20/30/50) wins customers back most profitably

Built for silent churn

A daily regular gone 10 days matters; a monthly one gone 10 days doesn't. A fixed “30 days” rule can't tell the difference.

Where the LLM fits

Gemini only writes the messages (first name only); Sarvam speaks them. Neither decides who or how much.

MODEL SELECTION

We tested it against simple rules — on real data

Why this model — tested against simple rules

Out of sample on both datasets · AUC = how well it ranks the customers who made no visits in the test period (0.5 = random, 1 = perfect) · reproduce: `npx tsx scripts/compare-models.ts`

MBG/NBD selected

Approach	Demo shop		CDNOW (real)	
	AUC	Forecast error	AUC	Forecast error
Average rate (rule)	0.76	4.05 visits	0.77	1.91 visits
Recency rule	0.89	–	0.54	–
BG/NBD	0.87	2.46 visits	0.32	0.78 visits
MBG/NBD	0.94	2.46 visits	0.69	0.76 visits
MBG/NBD · ranked by expected visits ✓	**0.95**	**2.46 visits**	**0.77**	**0.76 visits**
BG/NBD · ranked by expected visits	0.95	2.46 visits	0.77	0.78 visits

Plain BG/NBD assumes a one-time buyer is still a customer, which fails on real data where most people buy once (CDNOW AUC 0.32 — worse than random). MBG/NBD fixes that, and ranking by expected visits gives the best “who left” signal on both datasets while keeping forecast error less than half the simple rate rule's.

PROOF

Accurate on months it never saw

2,882

visits predicted

demo shop, trained to 1 Jun, tested 1 Jun – 15 Aug

2,899

visits that happened

–0.6% total error · per-customer r = 0.92

0.95

AUC · who left

demo shop · 0.77 on real CDNOW data

<3%

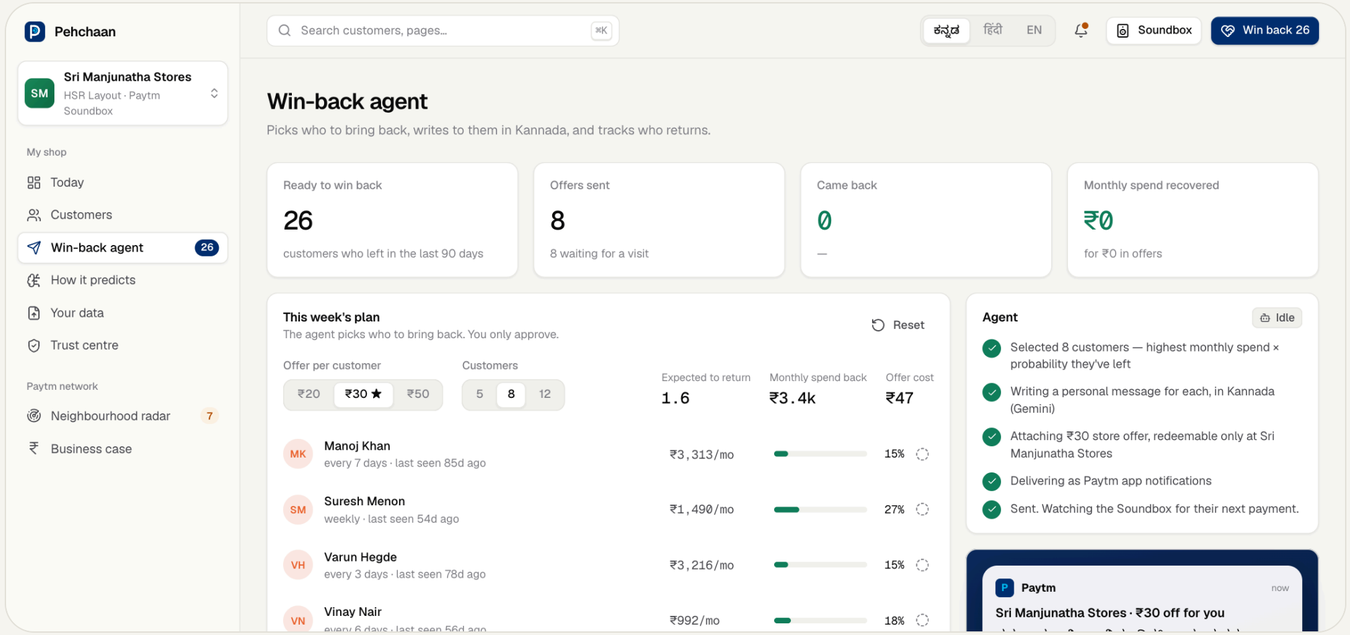
off the published paper

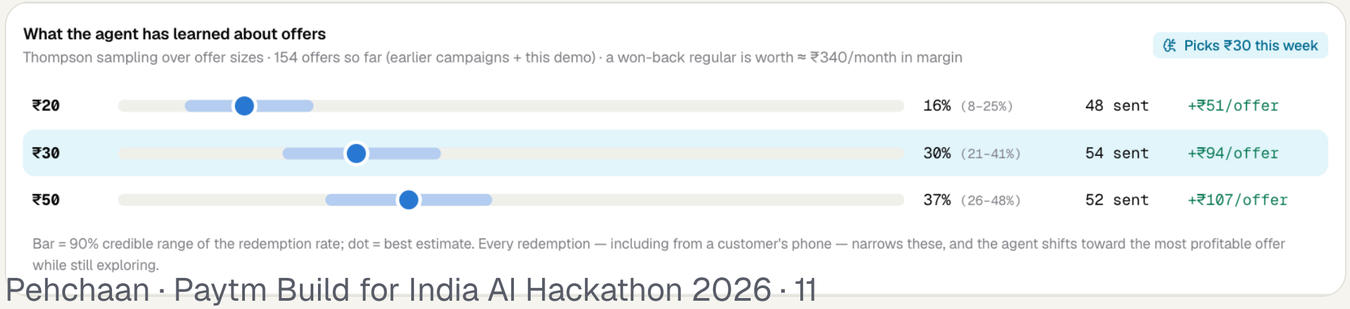
BG/NBD fit on real CDNOW purchases

Demo shop is a deterministic simulation (420 customers, 14,127 payments). Real-data check: the CDNOW sample from Fader, Hardie & Lee (2005) — our fit recovers r 0.242 / α 4.39 / a 0.81 / b 2.48 vs published 0.243 / 4.41 / 0.79 / 2.43. Reproduce: github.com/SayantikaLaskar/pehchaan

TRACK 3 · WIN-BACK AGENT

An agent that learns what works

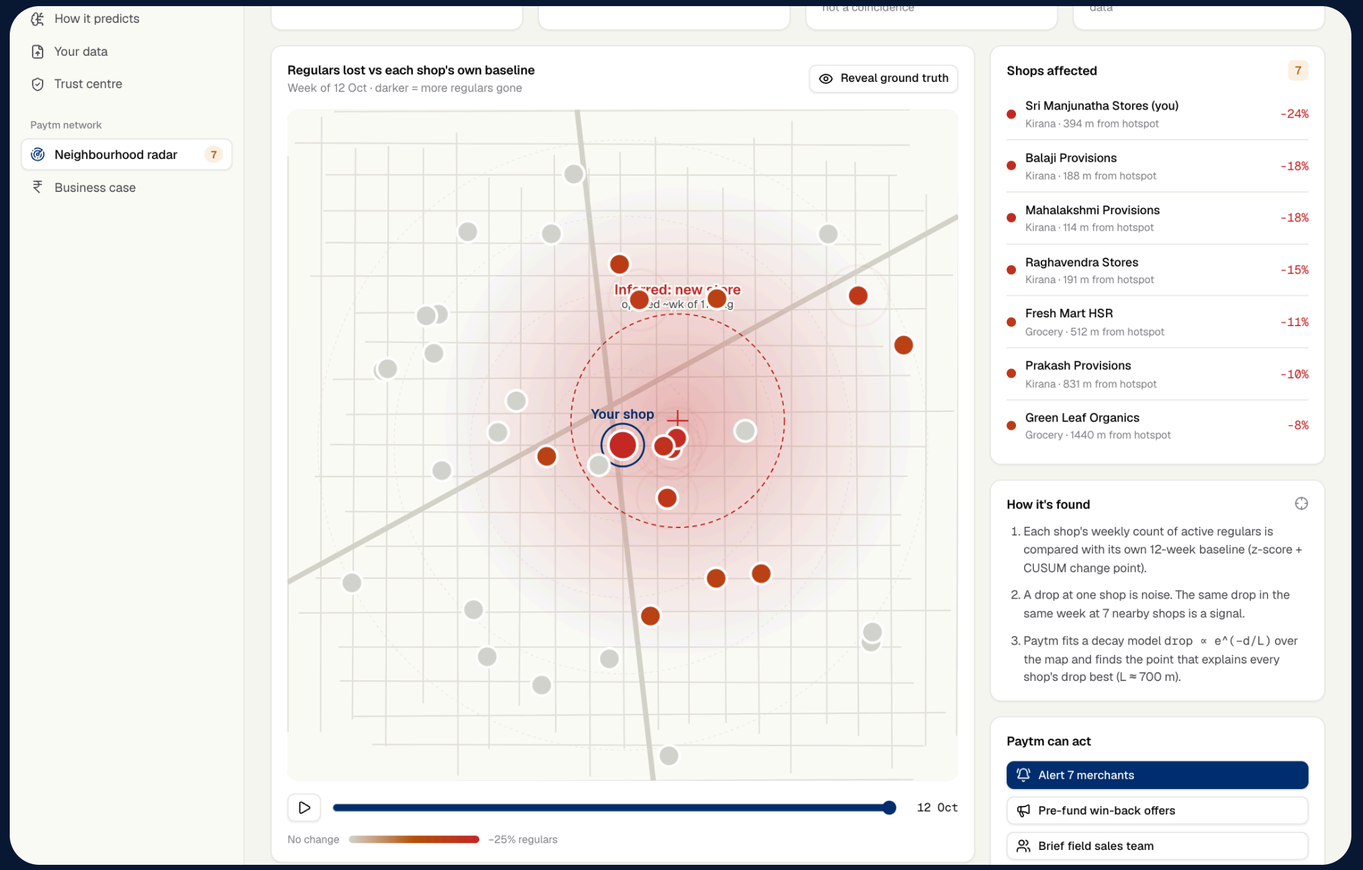




1. **Decides** who — only opted-in customers, max 1 offer / month
2. **Learns** the offer size by Thompson sampling
3. **Writes** in Kannada / Hindi / English
4. **Verifies** the return at the Soundbox and updates itself

NEIGHBOURHOOD RADAR · PAYTM VIEW

Paytm sees what no single shop can



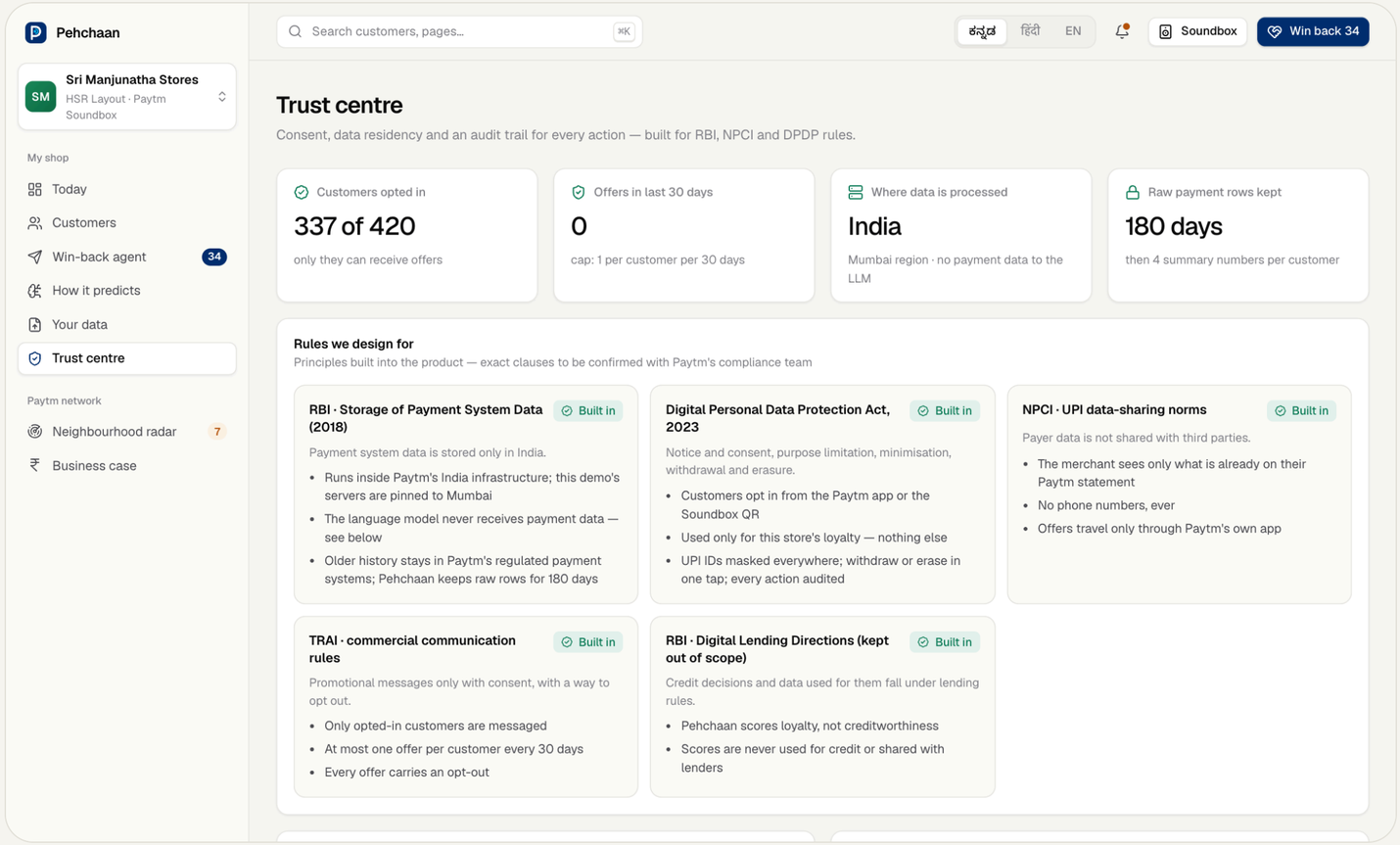
7 of 36
shops lost regulars in the same weeks

wk of 17 Aug
change point found — store opened 18 Aug

~360 m
error locating it, from aggregate drops alone

TRUST CENTRE

Built for RBI, NPCI and DPDP rules



- Payment data processed in India — servers in Mumbai
- LLM sees first name + offer only — never payment data
- Opt-in consent ledger, withdraw & erase, audit log
- One offer / month, opt-out on every message (TRAI)
- Loyalty only — never credit (RBI digital lending)

EDGE CASES

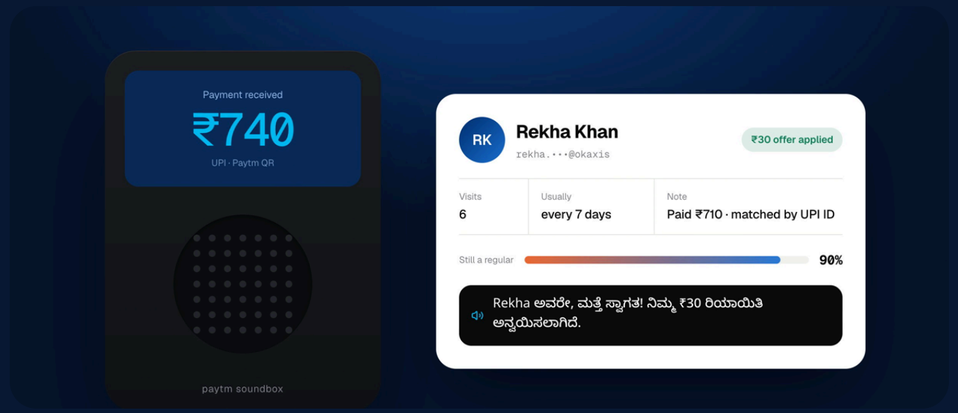
Built for the real India: lockdowns, outages, no signal

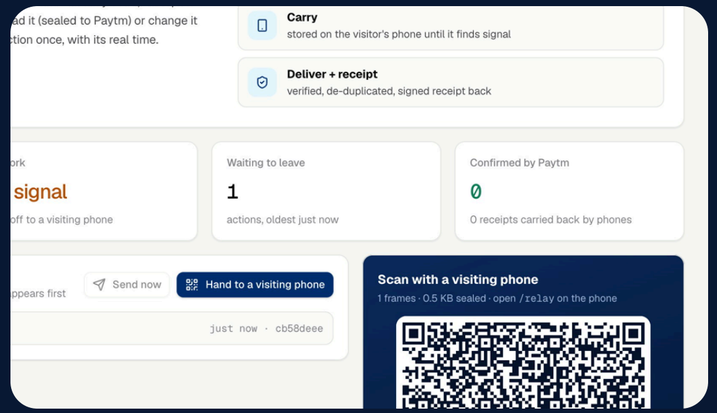
Situation	What Pehchaan does	Proof
Lockdown / city-wide shock	≥ 50% of shops drop together, no spatial source → **city-wide**. Churn clock switches to business time, discounts pause, agent moves to WhatsApp ordering.	False churn alarms **47 → 3** in the worst week
Soundbox / network outage	One shop at zero while peers trade → **outage**. Its scores freeze; payments backfill with real times.	False alarms **18 → 1**
Customer can't come in	Order on WhatsApp → priced, paid by UPI link, **counts as a visit**; the Soundbox announces the order.	₹0: wa.me + UPI links; Cloud API optional
Zero network at the shop	Offer stored against the **UPI ID** — customer just pays, Soundbox applies it. Data leaves on any visiting phone (Pehchaan Relay).	Relay: late, duplicate and tampered bundles tested

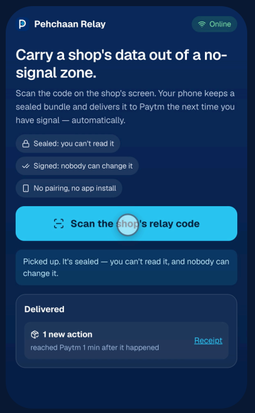
Paytm sees every merchant at once, so it can tell **why** payments stopped before telling a shopkeeper that customers left. Live: /app/resilience · /app/orders · reproduce: npm run resilience:check

PEHCHAAN RELAY · STORE, CARRY, FORWARD

Zero bars: the counter becomes its own network







No app, no data on the customer's side: the offer lives with the shop, keyed to the UPI ID.

01
Seal
AES-256-GCM to Paytm's key · ECDSA-signed by the shop device

02
Hand off
animated QR to any visiting phone — no pairing, no network

03
Carry
salesman, rider or customer; can't read or change it

04
Deliver
verified, applied once with its real time, signed receipt back

Tested: delivered 4 h late · second carrier → duplicate · tampered bundle → rejected (npm run relay:check). In production the same bundle also rides Bluetooth / Wi-Fi Direct (Nearby Connections) inside the Paytm app.

TECHNOLOGY

Tech stack

Data	Paytm QR / Soundbox payments — payer name + UPI ID; CSV import of real exports
Models	MBG/NBD + Gamma-Gamma (MLE, Nelder–Mead), Thompson sampling, CUSUM, spatial decay fit
Agent	Gemini Flash-Lite — structured JSON, cached, script-checked, template fallback
Voice	Sarvam AI — Bulbul v3 (kn / hi / en speech), Saaras v4 (spoken questions)
Live loop	Offer links + QR, phone page, events in MongoDB Atlas
App	Next.js 15, React 19, GSAP · Vercel Mumbai (bom1)

USP

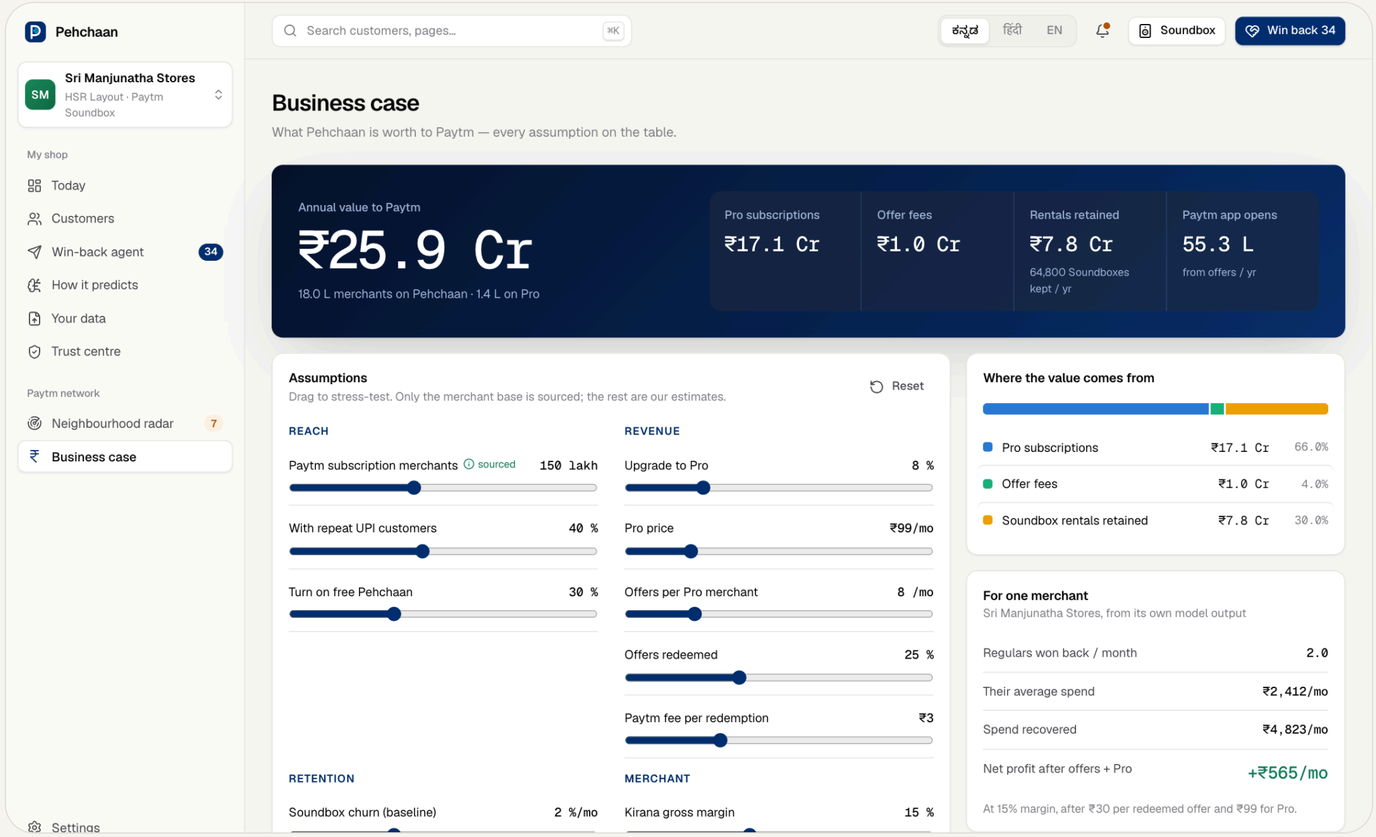
Why Pehchaan is different

Capability	Billing / loyalty apps	Generic AI chatbot	Pehchaan
Works with zero data entry	No	No	Yes
Predicts who is leaving	No	No	Yes
Acts on its own and learns	No	No	Yes
Speaks the owner's language at the counter	No	Yes	Yes
Sees neighbourhood-level shifts	No	No	Yes
Works with no network at the shop	No	No	Yes
Consent, residency, audit built in	No	No	Yes

The moat is Paytm's: the Soundbox at the counter, the consumer app to reach customers, and payment data across shops.

BUSINESS MODEL & IMPACT

₹26 Cr a year for Paytm, conservatively



Free → Pro ₹99/mo → take rate on offers

₹17.1 Cr Pro · ₹1.0 Cr fees · ₹7.8 Cr Soundbox rentals retained

One merchant (from its own model)

+₹565 / month net after offers + Pro

Only the merchant base (~1.5 Cr, Inc42) is sourced; every other input is an editable assumption in the live calculator.

ROADMAP

Risks, mitigations and an 8-week pilot

Privacy (DPDP)	No phone numbers; Paytm-app consent with opt-out; aggregated network views.
Cash customers	Only UPI regulars are scored — most repeat customers in a city kirana.
Shared phones / UPI IDs	Household grouping by surname, timing and basket; owner can merge or split.
Offer fatigue	Champions never get discounts; one offer / month; the agent learns.

Pilot · 50 kiranas, HSR Layout

Wk 1–2 — **Onboard & fit**
Soundbox opt-in; 6 months of real payments scored

Wk 3–6 — **Agent live**
Weekly win-backs; 20% of at-risk held out as control

Wk 7–8 — **Measure**
Return rate vs control, ₹ recovered per shop, retention

Success: return rate ≥ 2× control at positive ROI

Never lose a *regular* again.

Live demo · pehchaan-sigma.vercel.app
Code · github.com/SayantikaLaskar/pehchaan

Sayantika Laskar & Harsh Shukla
Thank you